Danial Al-Ghazali Walangadi

DKI Jakarta, Indonesia | danialagw@students.unnes.ac.id | +62-819-0215-7875 | agw2005.github.io

linkedin.com/in/agw2005 | github.com/agw2005

Research Statement

I am interested in the security-side of AI/ML and leveraging state-of-the-art AI technologies to address computer security challenges, including issues of privacy, accountability, and the sociotechnical aspects of cybersecurity.

Education

Universitas Negeri Semarang, BSc in Computer Science — 2023 – 2027 (expected)

- Overall GPA: 3.92/4.0 (Transcript)
- Major GPA: 3.92/4.0
- **Coursework:**

  Operating Systems (A), Algorithms and Programming (A), Introduction to Computer Science (A), Computer Architecture and Organization (AB), Computational Logic (A), Data Structures (A), Algorithm Analysis (A), Human-Computer Interaction (A), Computer Networks (A), Discrete Mathematics (A), Database Systems (A), Formal Languages and Automata Theory (A), Artificial Intelligence (A), Numerical Methods (A), Object-Oriented Programming (A), Image Processing (A), Software Engineering (A), Information Technology Research (AB), Compiler Design (AB).

Research Experience

System and AI Research Training Program (SYAIR) — Aug 2025 – Dec 2025

- Indonesia's international research training program instructed by **Prof. Haryadi Gunawi**, University of Chicago.
- Selected as one of the top 50 student researchers in the Computer Science field nationally.
- Reviewed more than 10 scholarly papers presented at leading computer science conferences, including but not limited to ACM CCS, NDSS, and EuroS&P.
- Analyzed codebases that are available from the supplementary materials of reproducible conference papers.
- Successfully replicated and thoroughly understood the full implementation of a conference paper on backdoor purification for malware classifiers using a fine-tuning method in the field of computer security.
  - Examined and reproduced critical experiments of "**PBP: Post-training Backdoor Purification for Malware Classifiers**" NDSS 2025. (Reproducibility Repository↗)
  - Studied, modified and replicated four crucial experiments of "**On the Detectability of ChatGPT Content: Benchmarking, Methodology, and Evaluation through the Lens of Academic Writing**" ACM-CCS 2024. (Reproducibility Repository↗)

Projects

Strengthened AES-128 Encryptor — Github↗

- Scaffolded a single-page web application that provides an implementation of AES-128 encryption alongside a reproduction of a research article that aims to strengthen the AES system by using novel S-Boxes.
- Designed the underlying AES-128 logic with TypeScript's standard library without any external libraries except Vitest as the testing suite.
- Overall source code length is over **2,500 LOCs** without counting Vite's React initialization overhead.
- The implementation was created with React, deployed with Cloudflare Pages, and is publicly available for use under the Apache 2.0 License.
- Tools Used: **TypeScript**, **React**, **TailwindCSS**, **Vitest**, **Cloudflare Pages**

Toy 16-bit RSA Encryptor — Github↗

- Built a simple web interface that provides an example of how RSA encryption works.

- Constructed the logic for generating RSA-relevant variables, RSA keys, and the encryptor-decryptor mechanism with TypeScript's standard library without any external libraries.
- Overall source code length is over **230 LOCs** without counting Vite's React initialization overhead.
- The implementation was created with React, deployed with Cloudflare Pages, and is publicly available under the Apache 2.0 License.
- Tools Used: **TypeScript**, **React**, **Cloudflare Pages**

License Plate Extractor Kaggle↗

- Created a deep learning model, allowing users to extract the text from images containing a license plate.
- The model was trained under Kaggle's T2 GPUs, with the codebase spanning over **740 LOCs**.
- The source code is available publicly under the 2.0 version of the Apache License.
- Tools Used: **PyTorch**, **YOLO**, **Pandas**, **Matplotlib**

Ngiprit Money Github↗

- Developed a web interface to keep track of transactions, debts, receivables, and their relevant statistics.
- The project comprises over **3,000 LOCs** without counting the overhead introduced by Laravel.
- The application can be used freely under version 3.0 of the GNU Affero General Public License.
- Tools Used: **Laravel**, **React**, **TailwindCSS**, **ChartJS**, **Date-fns**, **PostgreSQL**

Technical Skills

Languages: C++, Python, TypeScript, LaTeX

Machine Learning Frameworks: PyTorch

Cloud Platform: Digital Ocean, Chameleon Cloud

Embedded Systems Hardware: ESP32, Orange Pi

Web Development: React, TailwindCSS, Laravel

Database Systems: PostgreSQL, SQLite

Operating Systems: Windows, Debian WSL, Ubuntu

Language Proficiency

Indonesian Language
Native speaker

English December 2025
IELTS: 8.0 (C1)
L:9.0 ; R:9.0 ; W:7.0 ; S:6.5

Certification & Training

- Cisco Networking Academy Certified Computer Networking Skills (2024)
- Sun Asterisk Japanese Language Course (2025)

References

Prof. Haryadi Gunawi
Computer Science, University of Chicago
Research Program Instructor

Abas Setiawan, S.Kom., M.Cs.
Computer Science, Universitas Negeri Semarang
Lecturer, Academic Advisor